

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (CL) Examination

March - April 2018

CL421.01 Advanced Structural Analysis

Date: 06.04.2018, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Indian standard code IS-456:2000 and SP-16 are allowed.

SECTION – I

Q – 1 (a) Enlist the different uses of dome structure. [03]

(b) Define dome. Explain types of domes. [04]

Q – 2 (a) Enlist steps involved in the finite element analysis and explain any one in detail. [07]

Q – 2 (b) Determine shape functions for the constant strain triangle in terms of Natural Coordinate system. [07]

OR

Q – 2 Analyse the beam shown in **Figure 1** by finite element method. Find the unknown reactions. $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 4 \times 10^{-6} \text{ m}^4$. [14]

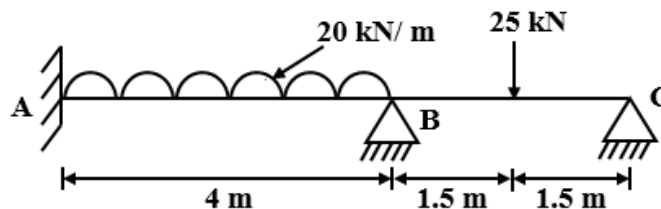


Figure: 1

Q - 3 Attempt any Two:

[14]

- (i) Describe advantages and disadvantages of Finite Element Method.
- (ii) Using Lagrange polynomial, find shape functions for the nine noded rectangular element in natural coordinate system where ξ and η are varying from -1 to 1.
- (iii) Assemble Jacobian matrix corresponding to the Gauss point (0, 0) for the element shown in **Figure 2**.

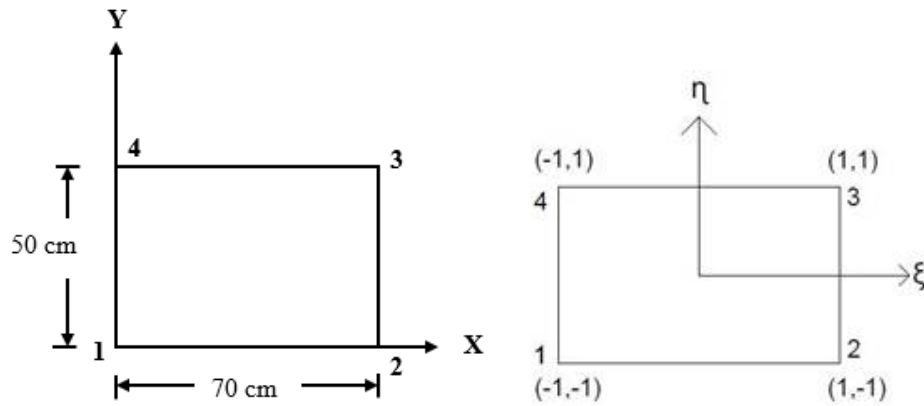


Figure: 2

SECTION – II

- Q – 4** Calculate combined joint load vector for the grid shown in **Figure 3**. Consider $EI = 10^7$ GJ. [07]

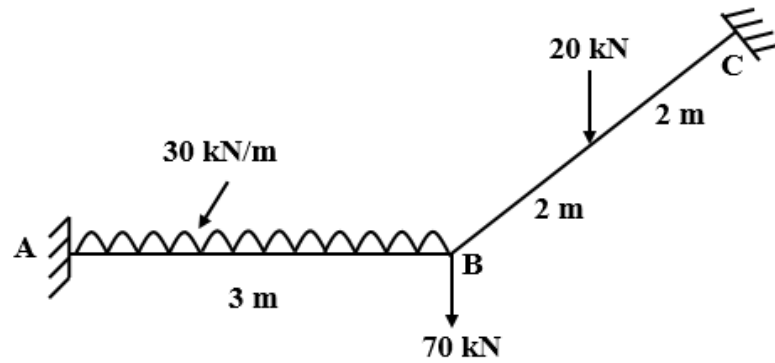


Figure: 3

- Q – 5** Develop member stiffness matrix in member axes for the space frame member with necessary diagrams. [14]

OR

- Q – 5** Analyse the space truss shown in **Figure 4** using stiffness method member approach. [14]
Calculate the forces in all members. Also draw axial force diagram. Consider $E = 2 \times 10^5$ N/mm² and $A = 1000$ mm² for all members.

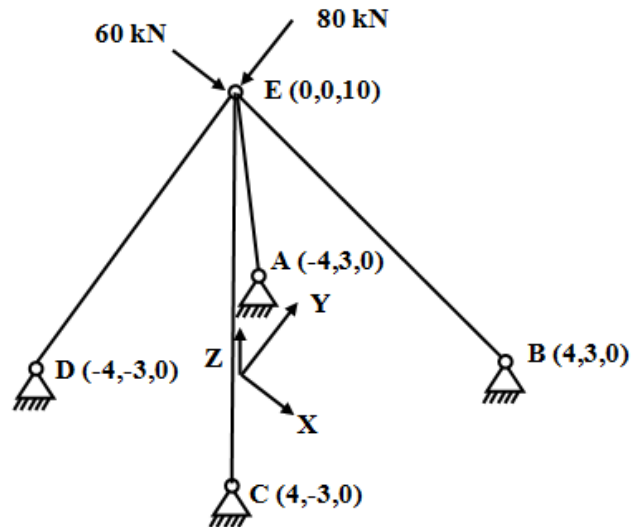


Figure: 4

Q – 6 (a) Explain following terms with specific diagrams wherever necessary. [05]

- | | | |
|---------------------|----------------------|-------------------------|
| (i) $[S_I]$ matrix | (ii) Elastic Support | (iii) Free Displacement |
| (iv) $[R_T]$ matrix | (v) Member axes | |

Q – 6 (b) A roof of a hall diameter 16 m is to be covered by a conical dome of 125 mm thickness and 4 m rise. The live load acting is 1000 N/m^2 . Calculate stresses in the dome and reinforcement required for the same. Use M 20 grade concrete and Fe 415 grade steel. [09]

$$\text{Meridional thrust } T = \frac{w}{2} \frac{y}{\cos^2 \theta}$$

$$\text{Hoop Force } H = w y \tan^2 \theta$$

OR

(b) Derive the equation of Meridional thrust and Hoop force for the spherical dome subjected to uniformly distributed load with neat sketch. [09]
